Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing

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<Water tank monitoring and automation system

Proposal Document

Group P1-4

IT Number	Name
IT24103508	Jayawardana M.D.V.L.
IT24101664	Jayasanka P.R.H.H.
IT24101931	Hettiarachchige G.D.P.
IT24102795	Alahakoon M.J.P.A.
IT24101275	Gamage H.B.G.S.D.
IT24103227	Weththewa S. N. W.

Background of study

This project is focused on developing water level monitoring and automation system. The main function of this is to detect the water level threshold of overflow level and lower level which can be considered as almost empty. When the water level reaches lower level, it automatically turns on the water pump and when it reaches to the overflow level, it automatically turns off. Water level can be monitored via LED bulb indicator on the tank and also through an application. This helps to prevent wastage of water and ease the process while avoiding any potential damages that can be happened.

Problem and Motivation

❖ Problem

In most of the households faces the issue regarding water wastage due to overflowing of water because of manual procedure. This leads to water loss, increase of utility bills, time wastage in order to monitor the water level. As the current procedure inefficient and consist with human error, it needs better solution.

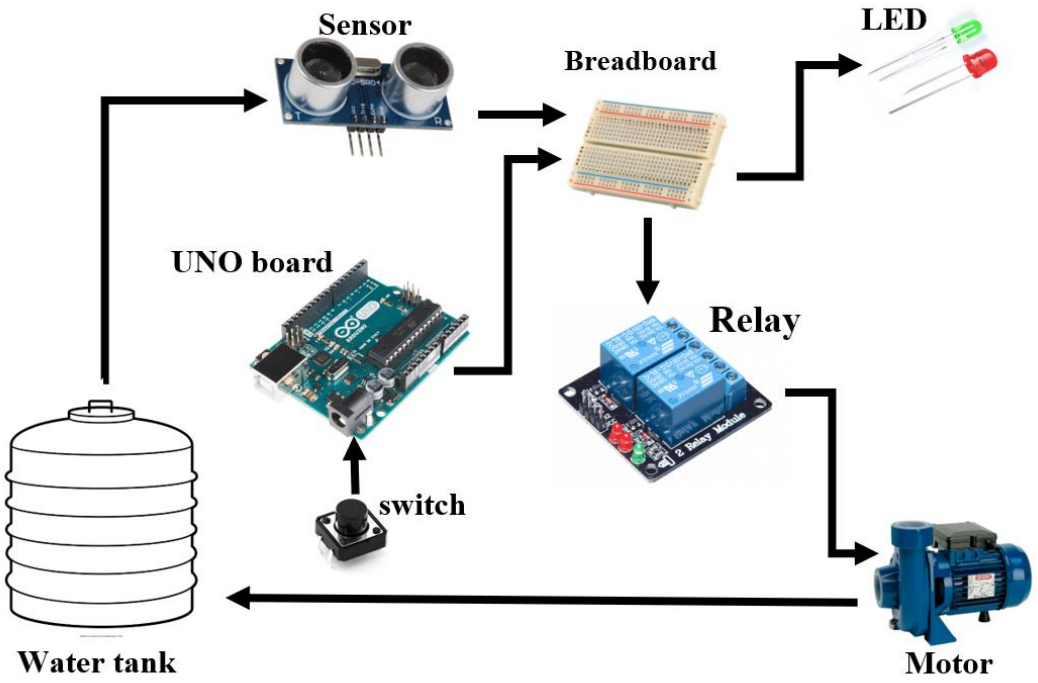
❖ Motivation

- Water conservation: The system controls automatically to prevent wastage of water and provides an optimum level of water.
- Property protection: It prevents overflow and various types of damage to property due to water by switching on the pump at the correct time.
- Cost effectiveness: Reduces Water Bills and Maintenance Costs Since it avoids excess use and reduces manual intervention to a minimum.
- Easy to use : Much of the water management is automated through automatic systems and real-time monitoring, which reduces manual oversight.

Aim and Objectives

- Aim
  - To create an automated system that detects and fills tanks efficiently, eliminating water waste and ensuring a consistent water supply.
- Objectives
  - Make a circuit with ultrasonic sensor that identifies the water level of the tank.
  - Make an LED bulb system that display water level on the tank
  - Use an automatic system to turn On/Off the motor which will fills the tank after reading systemized threshold levels

System Diagram



Methodology

❖ Component Selection

- The appropriate Arduino board must be connected. Therefore,
  - *HC-SR04 ultrasonic sensor*: Used for measuring the water level
  - *Relay Module*: Controls the on/off state of the water pump
  - *Water Pump*: Motor that fills the water tank

❖ Circuit Design

- The ultrasonic sensor is then connected to the Arduino through the Trig and Echo pins to the digital pins.
- Interface a relay module with the Arduino so that the water pump can be controlled.
- Connect and install a LED alongside the Arduino that will alert the user about the water level.

❖ Level Calculation

- Measure the distance from the sensor to the maximum level the water could fill and lower level which consider the level that should system starts refill the tank again.

❖ Notification System

- Run the Arduino program to make the LED on and sends related data on mobile phone about the water level

❖ Testing and Calibration of the System

- Test the system against accuracy over varying water levels.
- Calibrate the sensor readings if necessary

Evaluation Method

- ❖ Functional Evaluation
    - The accuracy of the sensors in reading the level of the water.
    - The system reports the level of water in real-time with accuracy.
    - The time taken in response to updating the level of the water by the system.
- ❖ Impact evaluation
    - Assessing whether improved water management has been enhanced, for instance, a reduction of water loss or preventing the overflowing of water in tanks.
    - Assess the contribution of the system towards low-flow or water conservation and sustainability goals
- ❖ Cost-Effectiveness
    - Assess whether the features provided justify the cost
    - Consider ongoing maintenance and power consumption
- ❖ User feedback
    - Gather feedback from users and evaluation

References

Dejan, n.d. *Ultrasonic Sensor HC-SR04 and Arduino – Complete Guide.* [Online]
Available at:
https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/&ved=2ahUKEwi-xrHQnOqHAxVF1DgGHa2-DWIQFnoECBoQAQ&usg=AOvVaw2qUPI0kBFWTYU9AhIz_8sA